```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

def run_nash_mattress_model():
    print("--- Loading Mattress Sales Mock Data ---")
    df = pd.read_csv("mattress_sales_mockup_3years.csv")
    df['Date'] = pd.to_datetime(df['Date'])

    # Aggregate weekly metrics across manufacturers and SKUs
    weekly_game = df.groupby('Date').agg({
        'Units_Sold': 'sum',
        'Effective_Price': 'mean',
        'Competitor_Avg_Price': 'mean',
        'Store_Foot_Traffic': 'mean',
        'Promo_Active': 'mean'
    }).reset_index()
    weekly_game = weekly_game.sort_values('Date').reset_index(drop=True)

    # Game-Theoretic Nash Equilibrium Metric:
    # Model price parity ratio (Our Effective Price vs Competitor Average Price)
    weekly_game['Price_Ratio'] = weekly_game['Effective_Price'] / weekly_game['Competitor_Avg_Price']

    # Nash Equilibrium Stability Index: Measures how close pricing is to competitive equilibrium (Ratio ~ 1.0),
    # adjusted for promotional response.
    weekly_game['Nash_Stability_Index'] = (1.0 - np.abs(weekly_game['Price_Ratio'] - 1.0)).clip(lower=0.0) * 100

    # Moving averages for trend clarity
    weekly_game['Rolling_Mean_Sales'] = weekly_game['Units_Sold'].rolling(window=12, min_periods=1).mean()
    weekly_game['Rolling_Mean_Nash'] = weekly_game['Nash_Stability_Index'].rolling(window=12, min_periods=1).mean()

    # Display console summary
    print(f"Total Weeks Analyzed: {len(weekly_game)}")
    print(f"Latest Weekly Sales Volume: {weekly_game['Units_Sold'].iloc[-1]} units")
    print(f"Latest Nash Equilibrium Stability Index: {weekly_game['Nash_Stability_Index'].iloc[-1]:.2f}%")
    print("\n--- Q4 Sales Trend Prediction & Competitive Game Dynamics ---")
    print("Interpretation: Nash Equilibrium measures market pricing stability and competitive parity.")
    print("During Q4, coordinated holiday discounting creates predictable strategic convergence across competitors.")

    # Plotting Results
    fig, axes = plt.subplots(2, 1, figsize=(12, 9), sharex=True)

    # Sales Volume Subplot
    axes[0].plot(weekly_game['Date'], weekly_game['Units_Sold'], label='Weekly Units Sold', color='#2b5c8f', alpha=0.5)
    axes[0].plot(weekly_game['Date'], weekly_game['Rolling_Mean_Sales'], label='12-Week Moving Average', color='#d95f02', linewidth=2.5)
    axes[0].set_ylabel('Mattress Unit Sales', fontsize=11)
    axes[0].set_title('Mattress Category Sales Trend & Q4 Forecasting Outlook (Nash Equilibrium Model)', fontsize=13, fontweight='bold')
    axes[0].legend(loc='upper left')
    axes[0].grid(True, linestyle='--', alpha=0.5)

    # Nash Equilibrium Stability Subplot
    axes[1].plot(weekly_game['Date'], weekly_game['Nash_Stability_Index'], label='Nash Equilibrium Stability (%)', color='#e7298a', linewidth=1.5, alpha=0.7)
    axes[1].plot(weekly_game['Date'], weekly_game['Rolling_Mean_Nash'], label='12-Week Moving Average (Stability)', color='#7570b3', linewidth=2)
    axes[1].axhline(y=weekly_game['Nash_Stability_Index'].mean(), color='red', linestyle=
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        ':', label='Mean Stability Baseline')
55      axes[1].set_ylabel('Market Stability Index (%)', fontsize=11)
56      axes[1].set_xlabel('Date', fontsize=11)
57      axes[1].set_title('Competitive Pricing Equilibrium & Strategic Parity (Game Theory
        Approach)', fontsize=13, fontweight='bold')
58      axes[1].legend(loc='upper left')
59      axes[1].grid(True, linestyle='--', alpha=0.5)
60
61      plt.tight_layout()
62      plt.savefig("mattress_nash_forecast.png")
63      print("Chart successfully saved to 'mattress_nash_forecast.png'.")
64
65  if __name__ == '__main__':
66      run_nash_mattress_model()
```